KQC7016 DATA ANALYTICS

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Lecturer: Associate Prof. Ir. Dr. Chow Chee Onn

Assignment 1

Github Link: https://github.com/U2103136/KQC7016-Data-Analytics/tree/main/Assignment%201

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1. Introduction

In smart agriculture, data is important for understanding the factors that influence crop yield. Crop production is affected by a combination of environmental conditions, soil characteristics, farming practices and crop health. Variables such as rainfall, soil moisture, humidity, pesticide usage, irrigation type, fertilizer type and crop disease status can help explain how crops perform under different growing conditions. Therefore, this assignment demonstrates how data analytics can support better decision-making in smart farming.

This study uses Exploratory Data Analysis (EDA) and statistical hypothesis testing to examine patterns in a smart farming dataset. EDA is used to understand distributions, categorical patterns, yield differences and relationships between variables. Statistical tests are then applied to determine whether the observed patterns are statistically meaningful rather than due to random variation.

The main objective is to identify whether environmental and farming conditions are related to crop yield and crop disease status. Specifically, this study examines: whether crop yield differs across environmental bands, whether environmental bands are associated with crop disease status, and whether fertilizer type and irrigation type have a combined effect on crop yield. This makes the analysis systematic and relevant to smart agriculture, where data-driven monitoring can support crop planning, disease management and farm productivity.

2. Description of Dataset and Preparation

The dataset used in this study contains both numerical and categorical agricultural variables. The numerical variables include soil moisture, soil pH, temperature, rainfall, humidity, sunlight hours, pesticide usage, crop duration, crop yield, location coordinates and NDVI index. These variables are suitable for descriptive statistics, distribution analysis, correlation analysis and yield comparison. The categorical variables include region, crop type, irrigation type, fertilizer type and crop disease status, which allow group-based comparisons.

The main outcome variable is `yield_kg_per_hectare`, which represents crop productivity. This variable is important because yield is a direct measure of agricultural performance. By comparing yield with environmental and management variables, the analysis can identify whether certain farming conditions are linked with higher or lower productivity.

The dataset contains 500 records and 22 columns. It includes observations from five regions: Central USA, East Africa, North India, South USA and South India. It also includes five crop types: maize, soybean, cotton, wheat and rice. This provides enough variety to compare crop yield patterns across different regions and crops.

Before analysis, the dataset was inspected for data types, missing values and duplicate records. No duplicate rows were found. The date columns, including `sowing_date`, `harvest_date` and `timestamp`, were converted into datetime format. A new variable, `growth_days`, was created by subtracting sowing date from harvest date. This represents the crop growth duration and was included in the numerical analysis.

Missing values were found in `irrigation_type` and `crop_disease_status`. Since these are categorical variables, the missing values were replaced with `"Unknown"` instead of using mean or median imputation. This allowed the records to remain in the dataset while still indicating that the information was not recorded. After cleaning, the dataset was ready for EDA and statistical testing.

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Table 1: Data Quality Summary

Metric	Value
Rows	500
Original Columns	22
Missing Values in irrigation_type	150
Missing Values in crop_disease_status	130
Duplicate Rows	0
Date Parse Errors	0
Sowing Date Range	2024-01-01 to 2024-03-28
Harvest Date Range	2024-04-09 to 2024-08-17
Timestamp Range	2024-01-03 to 2024-08-12

Table 2: Summary statistics of key numerical variables

Variable	Mean	Std. Dev.	Min	Median	Max
Soil Moisture (%)	26.750	10.150	10.160	25.855	44.980
Soil pH	6.524	0.586	5.510	6.530	7.500
Temperature (°C)	24.676	5.349	15.000	24.655	34.840
Rainfall (mm)	181.686	72.293	50.170	191.545	298.960
Humidity (%)	65.194	14.643	40.230	65.685	90.000
Sunlight Hours	7.030	1.692	4.010	6.995	10.000
Pesticide Usage (ml)	26.587	13.202	5.050	25.980	49.940
Growth Days	119.496	16.798	90.000	119.000	150.000
Yield (kg/ha)	4032.927	1174.433	2023.560	4071.690	5998.290
NDVI Index	0.602	0.175	0.300	0.610	0.900

3. Exploratory Data Analysis (EDA)

3.1 Distribution of Numerical Features

The distribution plots provide an overview of the main numerical variables in the dataset. Overall, the variables are spread across their expected ranges and no extreme visual outliers are obvious. Crop yield ranges from about 2,000 kg/ha to almost 6,000 kg/ha, showing clear variation in productivity. This variation is useful because it allows comparison of yield across different farming and environmental conditions

.Environmental variables such as rainfall, humidity, soil moisture and pesticide usage also show noticeable variation. This is important because these variables are later grouped into Low, Medium and High bands for statistical testing. NDVI index also varies across the dataset, which supports its use as a vegetation-health indicator. However, the distribution alone cannot confirm whether NDVI or any environmental variable strongly affects yield, so further correlation and statistical analysis are required.

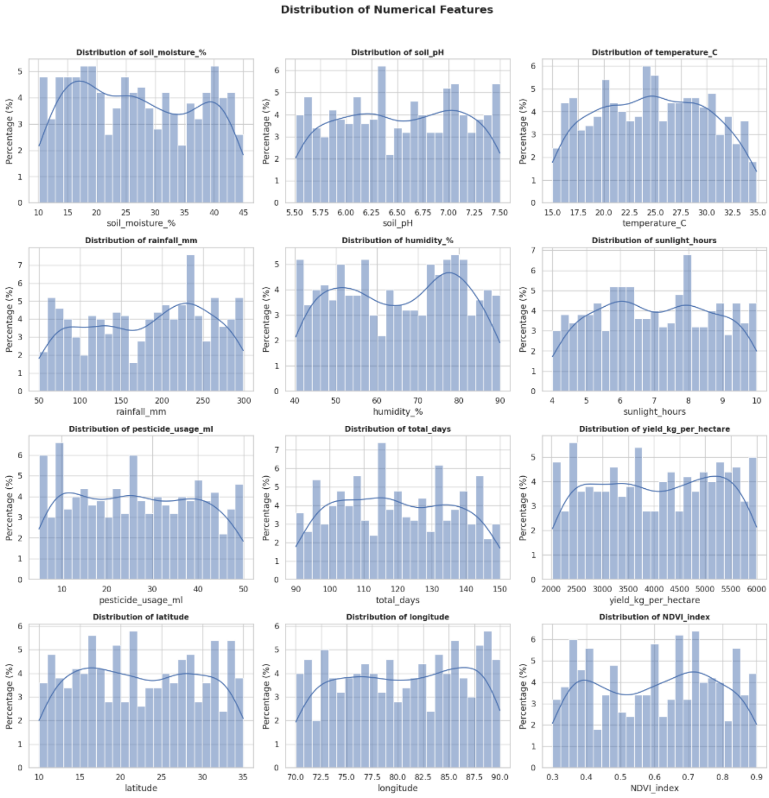

Figure 1: Distribution of Numerical Features

3.2 Categorical Feature Count Breakdown

The categorical plots show that the dataset contains a reasonable variety of crop types, fertilizer types, irrigation types and disease-status categories. Maize, soybean and cotton have slightly higher counts than wheat and rice, but all crop types are sufficiently represented. Fertilizer type is fairly balanced across inorganic, mixed and organic groups.

For irrigation type and crop disease status, some records are labelled as "Unknown" because missing values were retained as a separate category. This helps preserve the dataset size but also means that results involving these variables should be interpreted carefully. Overall, the categorical breakdown shows that the dataset is suitable for group comparison.

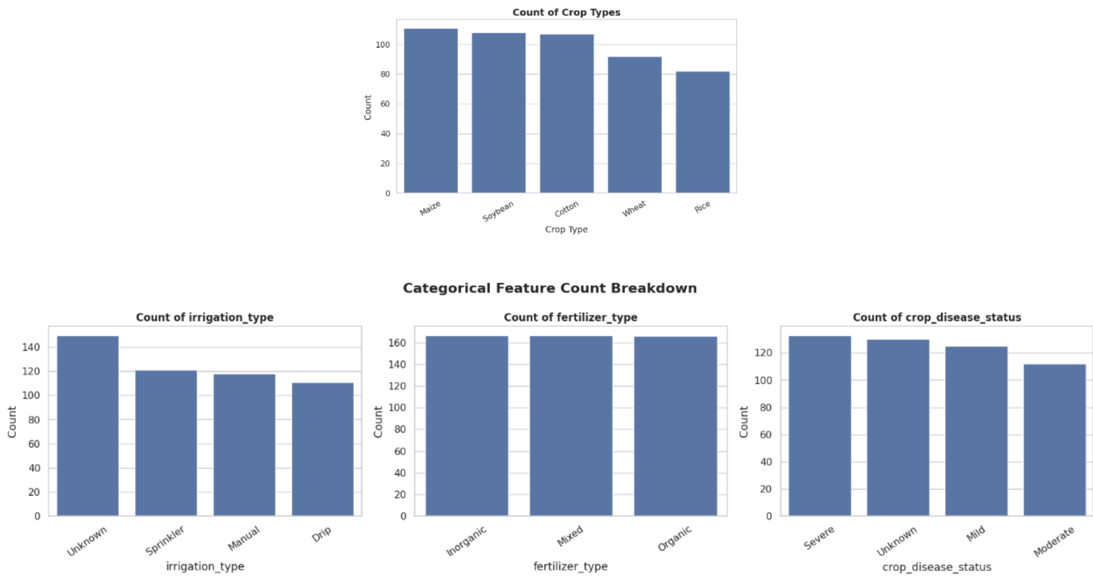

Figure 2: Categorical Feature Count Breakdown

3.3 Crop Disease Status by Region

The crop disease status chart shows that disease categories are not evenly distributed across regions. Some regions have higher counts of Severe or Mild cases, while others contain more Moderate or Unknown cases. For example, East Africa shows a relatively high number of Severe disease cases, while Central USA has high counts of Mild and Unknown cases.

This pattern suggests that crop health may vary by region. In smart agriculture, such information is useful because regions with higher disease counts may require closer monitoring, early warning systems or more targeted pesticide planning. However, this chart only shows frequency patterns, so statistical testing is needed to confirm whether environmental factors are associated with crop disease status.

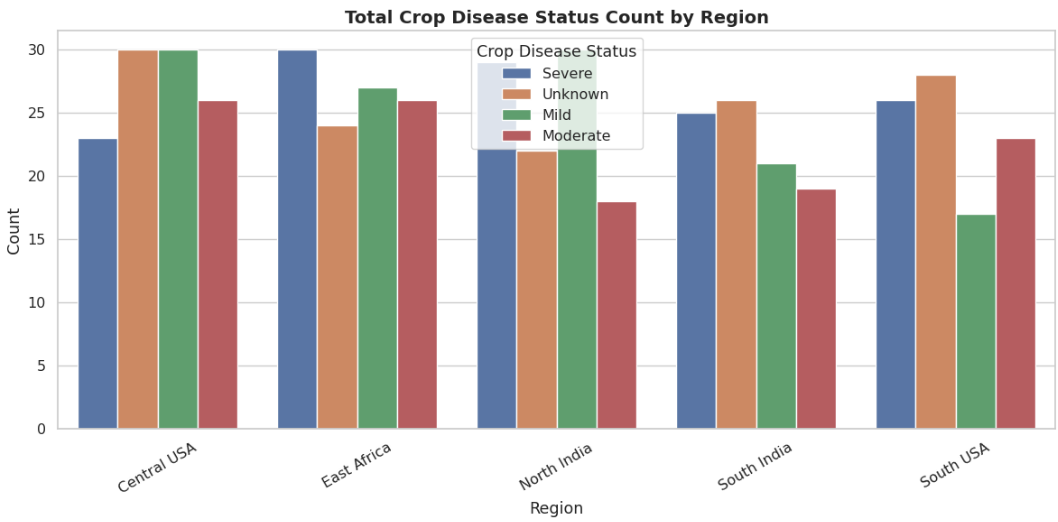

Figure 3: Total Crop Disease Status Count by Region

3.4 Crop Yield Distribution Across Categorical Features

The yield boxplots are one of the most important EDA outputs because crop yield is the main outcome variable. The plots compare yield across region, crop type, irrigation type, fertilizer type and crop disease status. Across regions, East Africa and South India appear to have slightly higher median yield, but the yield spread is wide in all regions. This suggests that region alone does not fully explain crop productivity.

For crop type, soybean appears to have a slightly higher median yield, while rice and cotton show more variation. However, the distributions overlap, meaning crop type alone may not strongly determine yield. Irrigation and fertilizer boxplots also show overlapping yield ranges. Drip irrigation and some fertilizer groups show slightly higher medians, but the differences are not visually strong enough to make a firm conclusion.

Overall, the boxplots suggest that yield is likely influenced by multiple factors rather than one single categorical variable. This supports the need for statistical testing, especially the Two-Way ANOVA between fertilizer type and irrigation type.

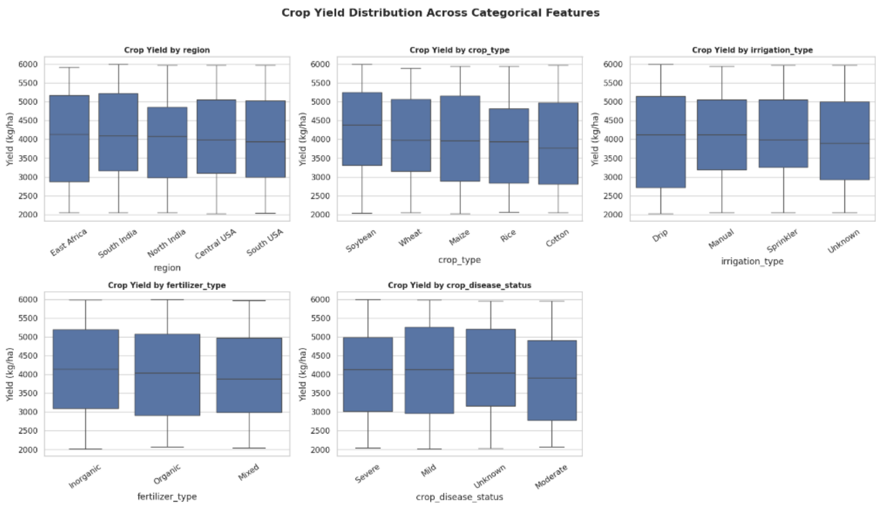

Figure 4: Crop Yield Distribution Across Categorical Features

3.5 Correlation Pattern Among Numerical Variables

The correlation heatmap shows mostly weak linear relationships between crop yield and the numerical variables. Crop yield has weak correlations with soil moisture, soil pH, temperature, rainfall, humidity, sunlight hours, pesticide usage, growth duration and NDVI index. This suggests that crop yield is not explained by one simple linear relationship.

However, weak correlation does not mean these variables are unimportant. In agriculture, environmental effects can be non-linear or range-based. For example, both too little and too much rainfall may affect crop performance. This is why rainfall, humidity, pesticide usage and soil moisture were later converted into Low, Medium and High bands for statistical testing.

The heatmap also shows a perfect correlation between total_days and growth_days, which is expected because both represent crop duration. In future modelling, one of them could be removed to avoid redundancy.

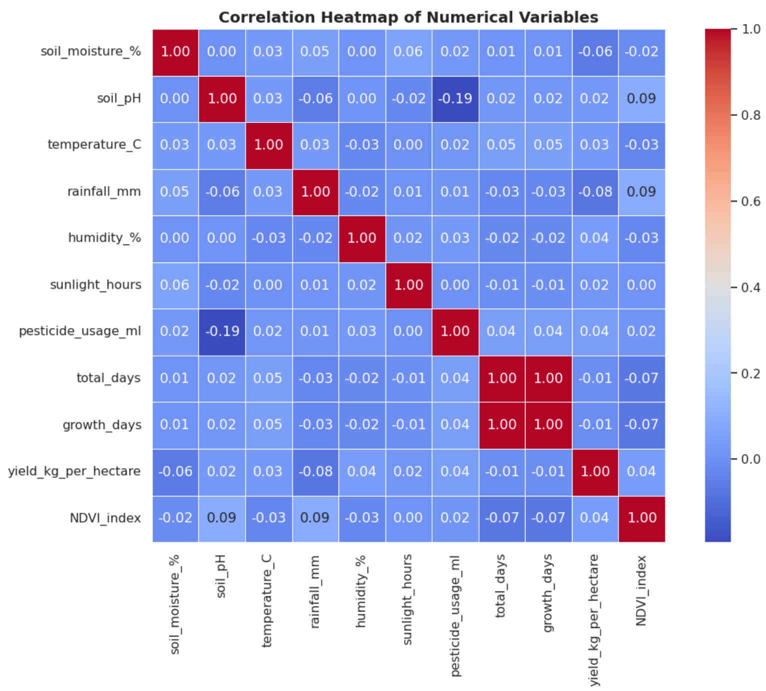

Figure 5: Correlation Heatmap of Numerical Variables

4. Statistical Analysis

The statistical analysis was conducted to confirm whether the EDA patterns were statistically meaningful. Three tests were used: One-Way ANOVA, Chi-Square Test of Independence and Two-Way ANOVA. A significance level of 0.05 was used. Therefore, p-values below 0.05 indicate statistically significant results, while p-values above 0.05 indicate insufficient evidence to reject the null hypothesis.

4.1 One-Way ANOVA: Environmental Bands and Crop Yield

One-Way ANOVA was used to test whether crop yield differs across Low, Medium and High bands of rainfall, humidity, pesticide usage and soil moisture. These variables were grouped into bands because agricultural conditions are often interpreted by ranges rather than raw values.

The results show that none of the environmental bands produced a statistically significant difference in crop yield. Rainfall band had a p-value of 0.2741, humidity band had a p-value of 0.3461, pesticide usage band had a p-value of 0.9175, and soil moisture band had a p-value of 0.1680. Since all p-values are above 0.05, crop yield does not differ significantly across these environmental bands.

Although soil moisture showed the clearest descriptive pattern, with the Low soil moisture band having the highest mean yield and the High soil moisture band having the lowest mean yield, the result was still not statistically significant. Therefore, the observed differences may be due to random variation. Since no ANOVA result was significant, Tukey HSD post-hoc testing was not required.

Table 3: One-Way ANOVA results for environmental bands and crop yield

Environmental Band	F-statistic	p-value	Decision
Rainfall Band	1.2976	0.2741	Not significant
Humidity Band	1.0624	0.3461	Not significant
Pesticide Usage Band	0.0861	0.9175	Not significant
Soil Moisture Band	1.7901	0.1680	Not significant

4.2 Chi-Square Test: Environmental Bands and Crop Disease Status

The Chi-Square Test was used to examine whether environmental bands are associated with crop disease status. This test is suitable because both variables are categorical. Unknown disease-status values were removed because they represent missing disease information.

Among the tested variables, only rainfall band showed a statistically significant association with crop disease status. The p-value was 0.0182, which is below 0.05. This means disease status appears to vary across rainfall levels. The Medium rainfall band had the highest number of Severe disease cases, suggesting that certain rainfall conditions may be linked with greater disease severity.

However, Cramer’s V was 0.1268, indicating a weak association. Therefore, rainfall may be useful as a disease-risk indicator, but it should not be treated as the only factor affecting crop disease. Humidity, pesticide usage and soil moisture bands were not significantly associated with crop disease status because their p-values were all above 0.05.

Table 4: Chi-Square test results for environmental bands and crop disease status

Environmental Band	Chi-square	df	p-value	Cramer's V	Decision
Rainfall Band	11.8889	4	0.0182	0.1268	Significant association
Humidity Band	0.4052	4	0.9821	0.0234	Not significant
Pesticide Usage Band	2.5155	4	0.6419	0.0583	Not significant
Soil Moisture Band	2.0064	4	0.7346	0.0521	Not significant

4.3 Two-Way ANOVA: Fertilizer Type × Irrigation Type on Crop Yield

Two-Way ANOVA was used to test whether fertilizer type, irrigation type and their interaction significantly affect crop yield. This is useful because water and nutrient management may work together to influence productivity.

The results show that fertilizer type was not statistically significant, with a p-value of 0.898. Irrigation type was also not significant, with a p-value of 0.917. The interaction between fertilizer type and irrigation type was not significant either, with a p-value of 0.445. Therefore, there is no strong evidence that fertilizer type, irrigation type or their combination significantly affects crop yield in this dataset.

The mean yield table shows some descriptive differences. For example, organic fertilizer with manual irrigation had the highest mean yield, while organic fertilizer with sprinkler irrigation had the lowest mean yield. However, because the interaction effect was not significant, these differences should be interpreted as descriptive patterns rather than confirmed effects.

Table 5: Two-Way ANOVA results for fertilizer type and irrigation type

Source	Sum Sq	df	F	p-value
Fertilizer Type	293698.882	2	0.107	0.898
Irrigation Type	235530.077	2	0.086	0.917
Fertilizer Type × Irrigation Type	5094389.686	4	0.933	0.445
Residual	46569566.095	341	NaN	NaN

Table 6: Mean yield by fertilizer type and irrigation type

Fertilizer Type	Drip	Manual	Sprinkler
Inorganic	4116.100	3982.474	4207.687
Mixed	3908.532	3998.669	4175.991
Organic	4050.036	4238.362	3857.727

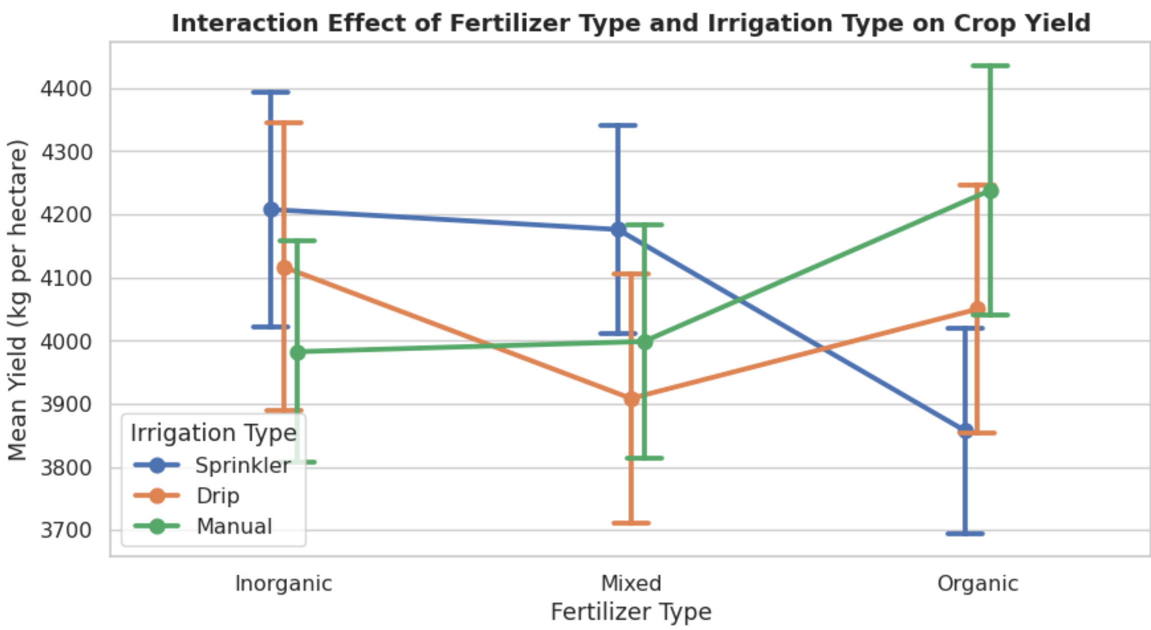

Figure 6: Interaction Effect of Fertilizer Type and Irrigation Type on Crop Yield

4.4 Assumption Checking

Assumption checking was conducted for the Two-Way ANOVA model. The Shapiro-Wilk test was used to check residual normality, while Levene's test was used to check homogeneity of variance.

The Shapiro-Wilk test produced a p-value of 2.261×10^{-9}, which is below 0.05. This indicates that the residuals are not normally distributed, so the ANOVA results should be interpreted cautiously. However, Levene's test produced a p-value of 0.5316, which is above 0.05. This means the equal-variance assumption is satisfied.

Overall, the assumption checks show that the Two-Way ANOVA is acceptable for exploratory interpretation, but the normality limitation should be acknowledged.

Table 7: Statistical assumption checks

Test	Statistic	p-value	Interpretation
Shapiro-Wilk Test	0.9511	2.261×10^{-9}	Residuals are not normally distributed
Levene's Test	0.8570	0.5316	Equal variance assumption is satisfied

5. Discussion

The results show that crop yield in the smart agriculture dataset is not controlled by one single factor. The EDA showed variation in crop yield across regions, crop types, irrigation methods, fertilizer types and disease categories. However, the correlation heatmap showed weak linear relationships between yield and most numerical variables. This suggests that crop productivity is complex and may depend on several conditions together rather than one direct relationship.

The One-Way ANOVA results support this interpretation. Rainfall, humidity, pesticide usage and soil moisture bands did not produce significant differences in crop yield. This does not mean these variables are unimportant, but it suggests that broad environmental bands alone cannot fully explain yield variation in this dataset. Soil moisture showed the clearest descriptive trend, but it was not statistically significant.

The Chi-Square test provided the strongest statistical finding. Rainfall band was significantly associated with crop disease status, although the relationship was weak. This suggests that rainfall may be useful for disease-risk monitoring, especially when combined with other crop-health indicators. The other environmental bands did not show significant associations with disease status.

The Two-Way ANOVA showed that fertilizer type, irrigation type and their interaction did not significantly affect crop yield. Although some fertilizer-irrigation combinations had higher mean yields, the differences were not statistically confirmed. This highlights the importance of not making recommendations based only on descriptive averages.

There are several limitations. First, the environmental bands were created using quantiles, not crop-specific agronomic thresholds. Second, some irrigation and disease-status values were recorded as Unknown, which may reduce interpretation accuracy. Third, the dataset does not include important variables such as soil nutrients, irrigation volume, fertilizer amount, crop variety or disease type. Future work could include these variables and develop predictive models for crop yield or disease risk.

6. Conclusion

This assignment shows that EDA and statistical testing can provide useful insights into smart agriculture data. EDA helped identify distributions, categorical patterns, crop disease differences, yield variation and correlation patterns. Statistical testing then confirmed which patterns were meaningful.

The One-Way ANOVA results showed that crop yield did not differ significantly across rainfall, humidity, pesticide usage or soil moisture bands. The Two-Way ANOVA also showed that fertilizer type, irrigation type and their interaction did not significantly affect crop yield. This suggests that crop yield is likely influenced by multiple interacting factors rather than one variable alone.

The strongest finding was from the Chi-Square test, where rainfall band was significantly associated with crop disease status. Although the effect size was weak, the result suggests that rainfall may be useful for crop health monitoring. In smart farming, rainfall information could support early warning systems when combined with other environmental and disease indicators.

Overall, the findings suggest that smart agriculture systems should monitor multiple factors together, including rainfall, soil moisture, disease status, irrigation method, fertilizer use and crop type. Future studies could improve the analysis by using expert-defined crop thresholds, adding more agronomic variables and building predictive models for yield or disease risk.

References

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